

1. Copy paste the folder you are going to work on to the laptop
2. Navigate into one stand/species folder
3. Create a new folder in this folder, name this folder: “MeasurementStandSpecies” (replace the Stand and Species with the ones in the folder name)
4. Species name: CANO, TSME, TSHE, THPL, PSME, ABAM; Stand ID: AV06, AB08, AO03, TO04, AE10, AG05, TA01, PARA, SUNR, AM16
5. You need to check the DPI value for each image by going into the folder-¿30per, then check the metadata.json. You will see a line with dpi.
6. Open Coorecorder, a window named ‘Setup project paths’ would pop up, if you don’t see it, go to File on the navigation bar, and then click on the Setup project paths.
7. For Project name, type in MORA2024StandSpecies (replace the Stand and Species with the folder name that you are going to work on)
8. For Image files directory tree, click on it and navigate to the folder you are going to work on
9. For .Pos files directory tree, click on it and navigate to the MeasurementStandSpecies folder you created
10. Click Ok & Close
11. Go to File on the navigation bar again, click on Open image file for new coordinates
12. Once the image opens, there will be a window pop up
- 13.
14. In the select measurement resolution section, make sure ‘Measure in milimetres’ is selected, then change the new DPI value to the value you have earlier from the metadata.
15. In the Data modes for dendrochronology section, make sure ‘Measure ring widths or generate Blue data or XRay density data. (which is usually the default)’
16. Click OK.

17. Now you can start adding points to the ring, if you don't remember how to do it, please refer to the tutorial video I sent to you.
18. Tip: Some rings might be wavy, use the multiple points to move around to find the most paralleled section
19. Tip: Some time when you try to open a file it might tell you unable to open the file, then just quit the application and open again, if it still doesn't work, leave a note
20. IMPORTANT: Always start from the very end of the core boundary (see the attached screenshots for some examples)
21. Only do the measurement when you are confident with it, if you are not sure or the scanning are unclear, you can either: (1) Go to the lab and refer to the physical cores; (2) Just leave a note and I will deal with them later
22. Once the measurement is done, go to File, click on Save As, name the .Pos file according to the folder name of this specific sample (E.g., PSME_AB08_4125(1))
23. When there are two samples for the same individual, finish the first one, save the measurement, and then click the CDendro icon in Coorecorder.
24. A window for the sample you just measured will pop up in CDendro, which you can use as a reference by clicking 'select as reference'.
25. Repeat step 10-20 for the second sample, but this time, try to save the sample every 20 rings and then click the CDendro icon.
26. Make sure the offset is 0, if not, it means you missed a ring.
27. Make sure all the red numbers are above 0.4, if not, there might be some misidentified rings.
28. Record the CoorC in the excel sheet when finish all the measurement for these two cores.
29. Fill out the excel sheet whenever one sample is done, record: how many rings measured, and notes need to pay attention later.